Transformations of Absolute-Value Graphs

Shifting the V, reflecting and stretching it, and reading the vertex off the equation

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- The parent graph $y = |x|$ is a V with its corner at the origin and arms of slope 1 and -1.
- In $y = a|x - h| + k$ the corner sits at (h, k). As always, $x - h$ moves the graph **right** by h — the bracket reads the opposite way round.
- a controls the arms: negative flips the V upside down, $|a| > 1$ makes it steeper, $|a| < 1$ makes it wider.
- The x-intercepts come from setting $y = 0$ and remembering that an absolute value equation splits into **two** cases; the y-intercept comes from putting $x = 0$.
- Where a grid is provided, plot the corner first, then the intercepts, then draw two straight arms.

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1. $f(x) = |x| - 3$

(a) Find the x-intercepts. (b) Find the y-intercept.

Answer: ________________

2. $f(x) = |x - 4|$

(a) Find the x-intercept. (b) Find the y-intercept.

Answer: ________________

3. $f(x) = |x+2| - 5$

(a) Find the x-intercepts. (b) Find the y-intercept.

Answer: ____________________

4. $f(x) = -|x| + 6$

(a) Find the x-intercepts. (b) Find $f(2)$.

Answer: ____________________

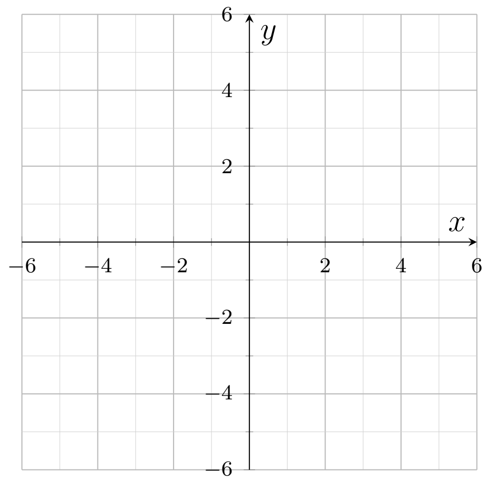
5. $f(x) = 2|x| - 8$

(a) Find the x-intercepts. (b) Find $f(3)$.

Answer: ____________________

6. $f(x) = |x - 1| - 4$

(a) Find the x-intercepts. (b) Plot the corner and the intercepts on the grid, then draw the graph.



Answer: ____________________

7. $f(x) = -2|x + 1| + 8$

(a) Find the x-intercepts. (b) Find the y-intercept.

Answer: ____________________

8. $f(x) = |x - 3| + 2$

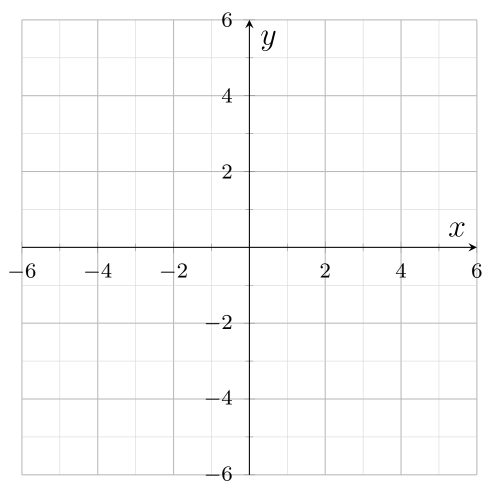
(a) Describe how this graph is obtained from the parent graph $y = |x|$. Name both moves and their directions.

(b) Find the value of f at the corner of the V, and put it on the answer line.

Answer: ________________

9. $f(x) = -|x + 2| + 3$

(a) Find the x-intercepts. (b) Plot the corner and the intercepts, then draw the graph.



Answer: ________________

10. $f(x) = 3|x - 2| - 6$

(a) Find the x-intercepts. (b) Find $f(5)$.

Answer: ____________________

11. A V-shaped graph has its corner at $(2, -3)$, opens upward, and its arms have slopes 1 and -1, just like the parent graph.

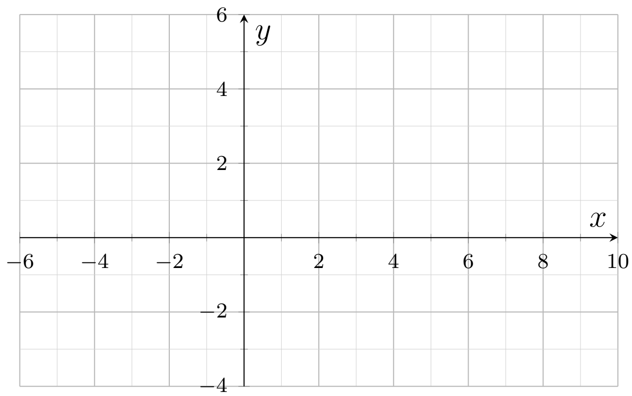
(a) Write the equation of this graph. (b) Find its x-intercepts.

Answer: ____________________

12. Synthesis challenge. $f(x) = -\frac{1}{2}|x-2|+3$

(a) Find the x-intercepts. (b) Find the y-intercept.

(c) Sketch the graph, then explain what the $\frac{1}{2}$ does to the arms and how that explains where the x-intercepts landed.



Answer: ______________